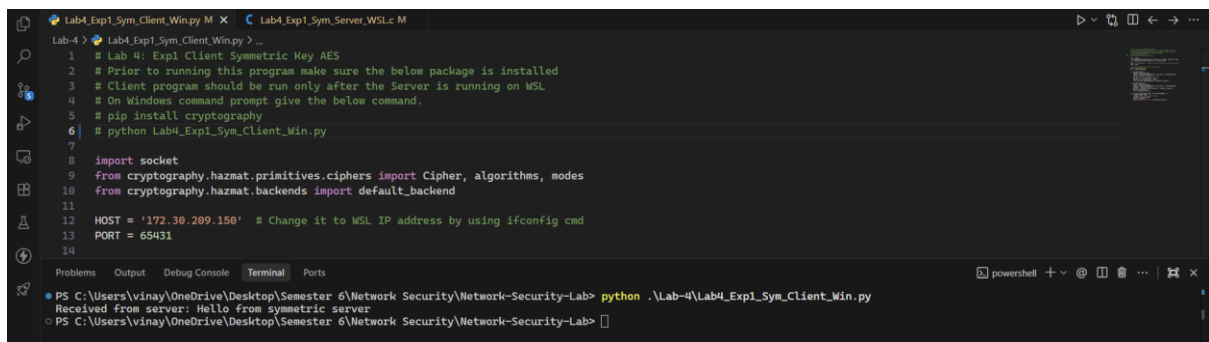
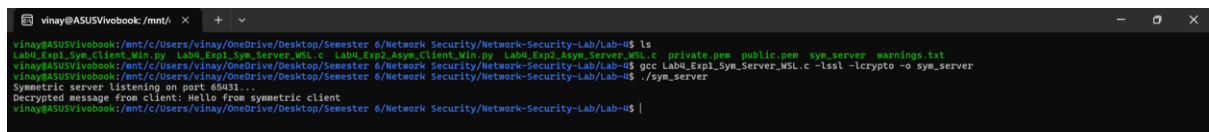


Lab-4-Exp-1 Client



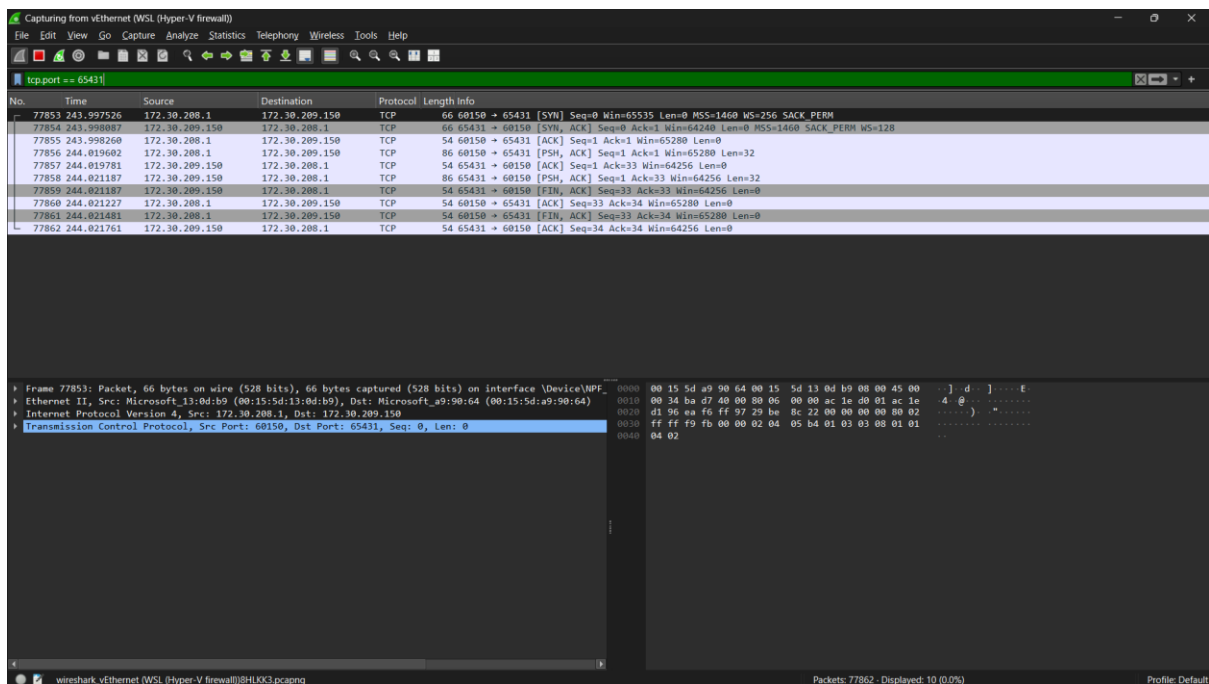
```
Lab4_Exp1_Sym_Client_Win.py M X Lab4_Exp1_Sym_Server_WSL.c M
Lab-4 > Lab4_Exp1_Sym_Client_Win.py > ...
1 # Lab 4: Exp1 Client Symmetric Key AES
2 # Prior to running this program make sure the below package is installed
3 # Client program should be run only after the Server is running on WSL
4 # On Windows command prompt give the below command.
5 # pip install cryptography
6 # python Lab4_Exp1_Sym_Client_Win.py
7
8 import socket
9 from cryptography.hazmat.primitives.ciphers import Cipher, algorithms, modes
10 from cryptography.hazmat.backends import default_backend
11
12 HOST = '172.30.209.150' # Change it to WSL IP address by using ifconfig cmd
13 PORT = 65431
14
Problems Output Debug Console Terminal Ports powershell + - @ [ ] ... [X] X
PS C:\Users\Vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab> python .\Lab-4\Lab4_Exp1_Sym_Client_Win.py
Received from server: Hello from symmetric server
PS C:\Users\Vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab>
```

Lab-4-Exp-1 Server



```
vinay@ASUSVivobook: /mnt/ > + -
vinay@ASUSVivobook: /mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ ls
Lab4_Exp1_Sym_Client_Win.py Lab4_Exp1_Sym_Server_WSL.c Lab4_Exp1_Sym_Client_Win.py Lab4_Exp1_Sym_Server_WSL.c
vinay@ASUSVivobook: /mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ gcc Lab4_Exp1_Sym_Server_WSL.c -lssl -lcrypto -o sym_server
vinay@ASUSVivobook: /mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ ./sym_server
Symmetric server listening on port 65431...
Decrypted message from client: Hello from symmetric client
vinay@ASUSVivobook: /mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ |
```

Lab-4-Exp-1 Wireshark



Capturing from vEthernet (WSL (Hyper-V firewall))

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 65431

No.	Time	Source	Destination	Protocol	Length	Info
77853	243.997526	172.30.208.1	172.30.209.150	TCP	66	60150 → 65431 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
77854	243.998067	172.30.209.150	172.30.208.1	TCP	66	65431 → 60150 [SYN, ACK] Seq=0 Ack=1 Win=64288 Len=0 MSS=1460 SACK_PERM WS=128
77855	243.998260	172.30.208.1	172.30.209.150	TCP	54	60150 → 65431 [ACK] Seq=1 Ack=1 Win=65280 Len=0
77856	244.019602	172.30.208.1	172.30.209.150	TCP	86	60150 → 65431 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=32
77857	244.019781	172.30.209.150	172.30.208.1	TCP	54	65431 → 60150 [ACK] Seq=1 Ack=33 Win=64256 Len=0
77858	244.021187	172.30.209.150	172.30.208.1	TCP	86	65431 → 60150 [PSH, ACK] Seq=1 Ack=33 Win=64256 Len=32
77859	244.021187	172.30.209.150	172.30.208.1	TCP	54	65431 → 60150 [FIN, ACK] Seq=33 Ack=33 Win=64256 Len=0
77860	244.021227	172.30.208.1	172.30.209.150	TCP	54	60150 → 65431 [ACK] Seq=33 Ack=34 Win=65280 Len=0
77861	244.021481	172.30.208.1	172.30.209.150	TCP	54	60150 → 65431 [FIN, ACK] Seq=33 Ack=34 Win=65280 Len=0
77862	244.021761	172.30.209.150	172.30.208.1	TCP	54	65431 → 60150 [ACK] Seq=34 Ack=34 Win=64256 Len=0

Frame 77853: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...

Ethernet II, Src: Microsoft_13:0d:b9 (00:15:5d:13:0d:b9), Dst: Microsoft_a9:90:64 (00:15:5d:a9:90:64)

Internet Protocol Version 4, Src: 172.30.208.1, Dst: 172.30.209.150

Transmission Control Protocol, Src Port: 60150, Dst Port: 65431, Seq: 0, Len: 0

0000 00 15 5d a9 90 64 00 15 5d 13 0d b9 00 00 45 00 ... d ... E

0010 00 34 ba d7 40 00 00 00 00 00 ac 1e d0 01 ac 1e ... 4 ...

0020 d1 96 ea 46 ff 97 29 be 8c 22 00 00 00 00 00 02 ...) ...

0030 ff f9 f9 00 00 02 04 05 b4 01 03 03 08 01 01

0040 04 02

Wireshark vEthernet (WSL (Hyper-V firewall))@HUK3.pcapng

Packets: 77862 - Displayed: 10 (0.0%)

Profile: Default

Lab-4-Exp2 Client

```
Lab4_Exp2_Asym_Server_WSL.c M X Lab4_Exp2_Asym_Client_Win.py M
Lab-4 > Lab4_Exp2_Asym_Server_WSL.c
1 // Lab 4: Exp2 Asymmetric Key RSA
2 // Before running the below command make sure the open SSL lib is installed on WSL
3 // sudo apt-get update
4 // sudo apt-get install libssl-dev
5 // Before running this program, generate both private.pem and public.pem on WSL side
6 // Use the below command for private.pem
7 // openssl genpkey -algorithm RSA -out private.pem -pkeyopt rsa_keygen_bits:2048
8 // Use the below command for public.pem
9 // openssl rsa -pubout -in private.pem -out public.pem
10 // Move the public.pem file from WSL to Windows directory where the client Python is running from
11 // Then, give the below command on WSL to build the executable
12 // gcc Lab4_Exp2_Asym_Server_WSL.c -lssl -lcrypto -o asym_server -Wno-deprecated-declarations
13 // Now run it using: ./asym_server first before running the client program on Windows
14

Problems Output Debug Console Terminal Ports
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab> python .\Lab-0\Lab4_Exp2_Asym_Client_Win.py
Received encrypted data from server: 0bfadee26cbc67dd8a8e0182c6589581ac1817c55636ee2992099f2738c8ab2ee427245945992a2bcc3e12dda0fc206b50f917814cab7a5b54229b5d9d5d8d3d1c583d729525c
e9f9634a7ee191d92eb1a7419666ecae4593debc59fe481e156d713833b1b828acdd2e9ecae7b4ec83c0bd00713471e4fd31e323ac5b0ea93bbe72c1f9db0badf94fdb71c6f1be17f9e798ba075a850eb26c890f8aaedf3a
e9beed1b747c12973fa83b9e5b58b581fa0316b9b1dd5cd4224fa4f1a863839d9fa48d6881d4ca15c6663214ac659b0e0e6c87329ed5943a41536d21eff458ff8c1721644bbfacc2f905cf65ec15724b959f83c2b45f8c9
f76c07c607c93
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab>
```

Lab-4-Exp-2 Server

```
vinay@ASUSVivobook:/mnt/ >
vinay@ASUSVivobook:/mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ ls
Lab4_Exp2_Asym_Client_Win.py Lab4_Exp2_Asym_Server_WSL.c Lab4_Exp2_Asym_Client_Win.py Lab4_Exp2_Asym_Server_WSL.c asym_server_warnings.txt
vinay@ASUSVivobook:/mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ openssl genpkey -algorithm RSA -out private.pem -pkeyopt rsa_keygen_bits:2048
.....
writing RSA key
vinay@ASUSVivobook:/mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ gcc Lab4_Exp2_Asym_Server_WSL.c -lssl -lcrypto -o asym_server -Wno-deprecated-declarations
vinay@ASUSVivobook:/mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$ ./asym_server
Asymmetric server listening on port 65432...
Decrypted message from client: Hello from asymmetric client
vinay@ASUSVivobook:/mnt/c/users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-Security-Lab/Lab-4$
```

Lab-4-Exp-2 Wireshark

```
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 65432

No. Time Source Destination Protocol Length Info
13 90.149206 172.30.209.1 172.30.209.150 TCP 66 60776 -> 65432 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14 90.150120 172.30.209.150 172.30.209.1 TCP 66 65432 -> 60776 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
15 90.150245 172.30.209.1 172.30.209.150 TCP 54 60776 -> 65432 [ACK] Seq=1 Ack=1 Win=65280 Len=0
16 90.150313 172.30.209.1 172.30.209.150 TCP 310 60776 -> 65432 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=256
17 90.150529 172.30.209.150 172.30.209.1 TCP 54 65432 -> 60776 [ACK] Seq=1 Ack=257 Win=64128 Len=0
18 90.150677 172.30.209.150 172.30.209.1 TCP 310 65432 -> 60776 [PSH, ACK] Seq=1 Ack=257 Win=64128 Len=256
19 90.150677 172.30.209.150 172.30.209.1 TCP 54 65432 -> 60776 [FIN, ACK] Seq=257 Ack=257 Win=64128 Len=0
20 90.150772 172.30.209.1 172.30.209.150 TCP 54 60776 -> 65432 [ACK] Seq=257 Ack=258 Win=65024 Len=0
21 90.150911 172.30.209.1 172.30.209.150 TCP 54 60776 -> 65432 [FIN, ACK] Seq=257 Ack=258 Win=65024 Len=0
22 90.160214 172.30.209.150 172.30.209.1 TCP 54 65432 -> 60776 [ACK] Seq=258 Ack=258 Win=64128 Len=0

Frame 13: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF... (AS
Ethernet II, Src: Microsoft_13:0d:b9 (00:15:5d:13:0d:b9), Dst: Microsoft_a9:9b:64 (00:15:5d:a9:9b:64)
Internet Protocol Version 4, Src: 172.30.209.1, Dst: 172.30.209.150
Transmission Control Protocol, Src Port: 60776, Dst Port: 65432, Seq: 0, Len: 0
0000 00 15 5d a9 9b 64 00 15 5d 13 0d b9 00 00 45 00 ... d ... E
0010 00 34 ba eb 40 00 00 00 00 ac 1e d0 01 ac 1e ... 4 ... @ ...
0020 d1 96 ad 68 ff 98 cb 3c 88 09 00 00 00 00 02 ... h ... x ...
0030 ff ff f9 fb 00 00 02 04 05 b4 01 03 08 01 01 ... ..
0040 04 02
```